EMBEDDED SYSTEMS PORTFOLIO

ESP32 • STM32 • FreeRTOS • Embedded Linux • Yocto Linux • IoT

What I Can Build

Custom firmware and embedded software for connected products, industrial devices, sensor systems, control electronics and IoT platforms. Projects can range from a single microcontroller feature to complete hardware-to-software integration.

Area	Capabilities
Microcontrollers	ESP32, STM32, FreeRTOS, peripheral configuration, bootloaders
Embedded Linux	Embedded Linux, Yocto Linux, system configuration, application integration
Drivers & Middleware	Sensor drivers, Linux drivers, HAL, APIs, communication middleware
Interfaces	UART, SPI, I2C, RS485, Modbus, CAN, Ethernet, GPIO, PWM
Wireless / IoT	WiFi, BLE, MQTT, cloud connectivity, remote monitoring
Sensors & Devices	NPK/soil sensors, industrial sensors, displays, meters, custom peripherals
Debug & Testing	Hardware bring-up, protocol debugging, firmware testing, integration

Typical Development Workflow

1. Requirements → 2. Hardware review → 3. Firmware/OS architecture → 4. Driver & peripheral development → 5. Integration → 6. Testing & debugging → 7. Documentation and delivery

Example Deliverables

- Clean C/C++ firmware with project structure and configuration.
- ESP-IDF / STM32 project setup and peripheral integration.
- FreeRTOS tasks, queues, timers, event handling and communication stacks.
- Yocto Linux layers/recipes and application integration where required.
- Sensor, display, communication and industrial protocol integration.
- Debugging support for hardware/firmware integration issues.
- Build instructions, configuration notes and handover documentation.

Platforms & Technologies

Platforms: ESP32, STM32 and embedded Linux-capable SoCs
Software: C, C++, Python, FreeRTOS, Embedded Linux, Yocto Linux
Connectivity: WiFi, BLE, Ethernet, MQTT

Industrial: RS485, Modbus, CAN
Hardware: Sensors, displays, custom PCBs, power/control interfaces

Need a custom embedded product?

Share your MCU/SoC, schematic or datasheet, required features, interfaces and existing code (if any). The project can then be scoped around your exact hardware and software requirements.